Diagonalization

Recall from last class:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v, called an **eigenvector** corresponding to λ, such that
$$Av = \lambda v.$$
- If λ is an eigenvalue of an $n \times n$ matrix A, then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ.
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \dots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_p$ of an $n \times n$ matrix A, then $v_1, \dots, v_p$ are linearly independent.
- If A is an $n \times n$ matrix, then the characteristic polynomial of A, denoted by $p_A(\lambda)$, is given by
$$p_A(\lambda) = \det(A - \lambda I_n).$$
- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).

1. Let $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$. Find eigenvalues and eigenspaces of A.

Definition 1. *An $n \times n$ matrix A is* **diagonalizable** *if A is similar to a diagonal matrix.*

Diagonalization Theorem: An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A. In this case the diagonal entires of D are the corresponding eigenvalues of A.

Definition 2. *A basis of* $\mathbb{R}^n$ *diagonalizing* A *is called an* **eigenbasis***.*

2. Prove the Diagonalization Theorem.

3. Let $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ If possible, diagonalize A.

4. Let $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ If possible, diagonalize A.

Theorem 6: An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.

5. Prove the theorem.

Diagonalization– Answers / Solutions

1. In order to find the eigenvalues, we compute the zeros of the characteristic polynomial. Note $p_A(\lambda) = \det \begin{bmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{bmatrix} = (1-\lambda)(4-\lambda)+2 = (\lambda-2)(\lambda-3)$. Thus the eigenvalues of A are 2 and 3. We find the eigenspaces for each now.

$\underline{\lambda = 2}$. In this case, we get

$$A - 2I_2 = \begin{bmatrix} -1 & 1 \\ -2 & 2 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Thus the eigenspace corresponding to the eigenvalue of 2 is given by

$$E_2 = N(A - 2I_2) = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

$\underline{\lambda = 3}$. In this case, we get

$$A - 3I_2 = \begin{bmatrix} -2 & 1 \\ -2 & 1 \end{bmatrix} \sim \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}.$$

Thus the eigenspace corresponding to the eigenvalue of 3 is given by

$$E_3 = N(A - 3I_2) = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}.$$

Thus we get

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^{-1}.$$

2. *Proof.* We prove the reverse implication first.

$\Longleftarrow$. Let $v_1, \ldots, v_n$ be linearly independent eigenvectors with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$. Define $P = [v_1 \ \ldots \ v_n]$ and $D = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 & 0 \\ 0 & \lambda_2 & \ldots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & \lambda_n \end{bmatrix}$. Then

$$\begin{aligned} AP &= A[v_1 \ \ldots \ v_n] \\ &= [Av_1 \ \ldots \ Av_n] \\ &= [\lambda_1 v_1 \ \ldots \ \lambda_n v_n] \\ &= PD. \end{aligned}$$

Since P is invertible, we get $A = PDP^{-1}$, and hence A is diagonalizable.

$\Longrightarrow$. Suppose $A = PDP^{-1}$ for some matrix P and some diagonal matrix D. Let $P = [v_1 \ \ldots \ v_n]$ and let D be diagonal matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$. Since P is invertible, the Invertible Matrix Theorem gives $v_1, \ldots, v_n$ are linearly independent. Thus we only need to prove that $v_1, \ldots, v_n$ are eigenvectors of A. Let $e_1, \ldots, e_n$ denote the standard basis vectors for $\mathbb{R}^n$, then note $v_k = Pe_k$ and hence $P^{-1}v_k = e_k$. Thus

$$Av_k = PDP^{-1}v_k = PDe_k = P\lambda_k e_k = \lambda_k Pe_k = \lambda_k v_k.$$

Thus $v_1, \ldots, v_k$ are eigenvectors of A with eigenvalues $\lambda_1, \ldots, \lambda_k$. $\square$

3. In order to find the eigenvalues, we compute the zeros of the characteristic polynomial. Note $p_A(\lambda) = \det \begin{bmatrix} 1-\lambda & 3 & 3 \\ -3 & -5-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{bmatrix} = -(\lambda-1)(\lambda-2)^2$. Thus the eigenvalues of A are 1 and -2 (with algebraic multiplicity 2). We find the eigenspaces for each now.

$\underline{\lambda = 1}$. In this case, we get

$$A - I_3 = \begin{bmatrix} 0 & 3 & 3 \\ -3 & -6 & -3 \\ 3 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus the eigenspace corresponding to the eigenvalue of 1 is given by

$$E_1 = N(A - I_3) = \text{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

$\underline{\lambda = -2}$. In this case, we get

$$A + 2I_3 = \begin{bmatrix} 3 & 3 & 3 \\ -3 & -3 & -3 \\ 3 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus the eigenspace corresponding to the eigenvalue -2 is given by

$$E_{-2} = N(A + 2I_3) = \text{Span}\left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Thus we get

$$A = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}.$$

4. In order to find the eigenvalues, we compute the zeros of the characteristic polynomial. Note $p_A(\lambda) = \det \begin{bmatrix} 2-\lambda & 4 & 3 \\ -4 & -6-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{bmatrix} = -(\lambda-1)(\lambda-2)^2$. Thus the eigenvalues of A are 1 and -2 (with algebraic multiplicity 2). We find the eigenspaces for each now.

$\underline{\lambda = 1}$. In this case, we get

$$E_1 = N(A - I_3) = \text{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

$\underline{\lambda = -2}$. In this case, we get

$$E_{-2} = N(A + 2I_3) = \text{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Hence we have only two linearly independent eigenvectors and by the Diagonalization theorem, A cannot be diagonalized.